

3D Printable Play-Dough

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Figure 1: A set of play-dough prints made from different recipes.

ABSTRACT

Play-dough is a brightly-colored, easy-to-make, and familiar material. We have developed and tested custom play-dough materials that can be employed in 3D printers designed for clay. We demonstrate how custom-color prints can be designed and constructed. We also explore the design potential of play-dough as a sustainable fabrication material, highlighting its recyclability, compostability, and repairability. Finally, we present a set of example artifacts made from play-dough. We believe play-dough 3D printing can enable new kinds of aesthetic and functional applications in digital fabrication and could make 3D printing more feasible in a range of educational settings.

CCS CONCEPTS

• Computer systems organization → Embedded systems.

KEYWORDS

3D printing, digital fabrication, biodegradable, compostable

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1 INTRODUCTION

Play-dough is cheap, easy-to-make, biodegradable, and can be dyed with food coloring into vivid colors. We have developed workflows that allow it to be printed on low cost desktop clay 3D printers, like the Eazao Zero, which sells for under \$1000 USD. There has been some very limited exploration of the 3D printing of play-dough in the past. In particular, the 3D printing company Hyrel has demonstrated that it is possible to print this material [4]. However, there is no research literature on this topic and the creative potential of play-dough has not been explored.

We introduce recipes and workflows for 3D printing home-made play-dough. We investigate the printability of several different play-dough recipes, print a range of geometry with play-dough and compare the material's sustainability to PLA (polylactic acid).

The use of play-dough in 3D printing opens up a range of new opportunities which we also begin to explore. In particular, users can easily generate their own custom-color filaments from play-dough and print in full-color. Our software supports a low-resolution mapping of color to computationally generated features of form, similar to the ikat method of dying yarn before weaving [3].

2 SOFTWARE

To generate 3D printer paths to evaluate our materials, we wrote a custom open-source software library that enables us to create paths through space and then translate these paths into g-code, the file-format that defines the toolpath of 3D printers [6]. It is intended to give programmers fine-grained control over the surface texture of 3D prints as well as their geometry. The Python-based library is employed in the context of Grasshopper and Rhino and employs Rhino to generate 3D visualizations of tool-paths and prints. Figure 2 shows how our software lets users add texture and color to imported mesh geometry. More information about the software can be found in [1, 6].

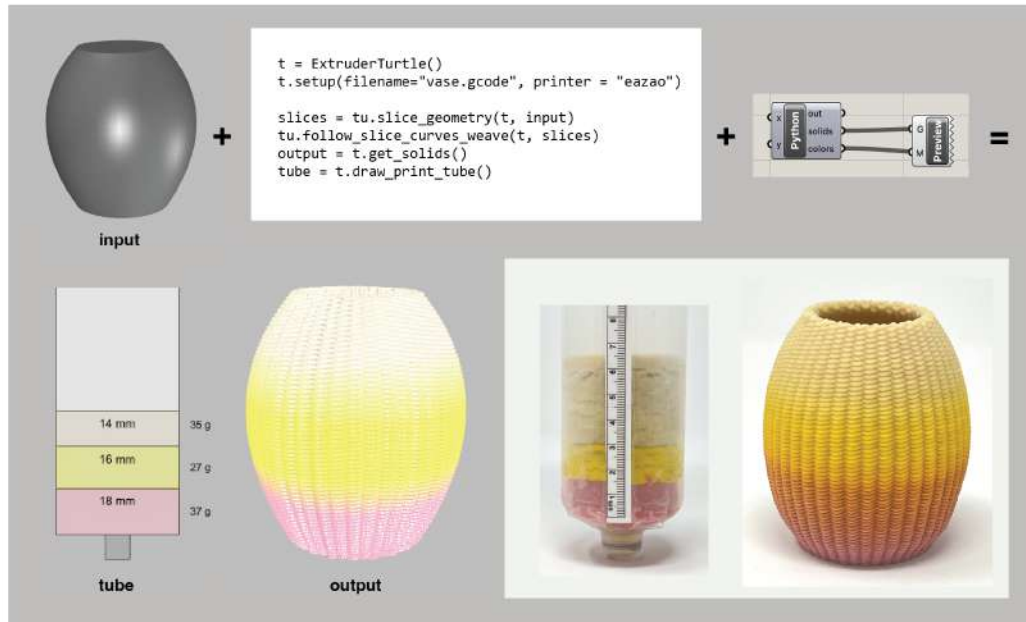


Figure 2: Our software. Top: An input shape and code snippet. Bottom: The output of the code is a visualization of the printer tube and toolpath. The actual print tube and printed object is also shown on the right.

Traditional	Corn	Corn and Wheat	Glue and Sand	25% Clay
200g wheat flour 100g salt 15g cream of tartar 30g vegetable oil 200g water	200g corn flour 30g vegetable oil 150g vinegar	150g corn flour 50g wheat flour 30g vegetable oil 150g vinegar	150g corn flour 50g wheat flour 200g play sand 50g white glue 100g water	by weight: 75% corn & wheat play-dough 25% stoneware clay

Table 1: Play Dough Recipes

3 RECIPES

Table 1 lists the five play-dough recipes we explored. 1) A traditional wheat flour and salt-based dough labeled Traditional (column 1), 2) A 100% gluten free dough made with corn flour labeled Corn (column 2), 3) A dough made from 75% corn flour and 25% wheat flour labeled Corn and Wheat (column 3), 4) A dough made from 75% corn flour and 25% wheat flour that is strengthened with the addition of white glue and sand, and 5) A dough made from 75% Corn and Wheat play-dough (from our recipe) and 25% stoneware clay.

For the traditional recipe, all ingredients are mixed together and then cooked over medium heat until the dough is at the desired hardness. For all other recipes, the ingredients are mixed together in a kitchen mixer. The oil should be added after the other ingredients have been mixed slightly to prevent clumping. No cooking is required. Once thoroughly mixed, water can be added to achieve the desired hardness. Figure 1 shows an image of bowls printed from each recipe. Based on this experiment and others, described in detail in [2], we determined that the Glue and Sand recipe is a

viable general purpose 3D printing material. The Corn and Wheat recipe can be used to print simple vessels with walls of uniform thickness, but the material is prone to cracking when used to print more complex geometries. The other materials were not viable 3D printing substrates.

4 SUSTAINABILITY CONSIDERATIONS

An important advantage of play-dough in comparison to PLA is its relative sustainability. The ingredients for our play-doughs are all renewable. Play-dough is easily recyclable. Dried prints and waste can be dissolved in water and then re-printed or mixed into new batches of dough. Prints made from play-dough are also very easy to repair with water-based craft glue. Figure 4 top shows the process of gluing a broken piece back onto a play-dough print. Because prints are so porous, repair seams are virtually undetectable.

Finally, play-dough prints are compostable. Figure 4 bottom show an image of 3D printed play-dough cylinders that were left in a moist sealed tupperware at room temperature for one month, demonstrating how quickly they naturally decay.

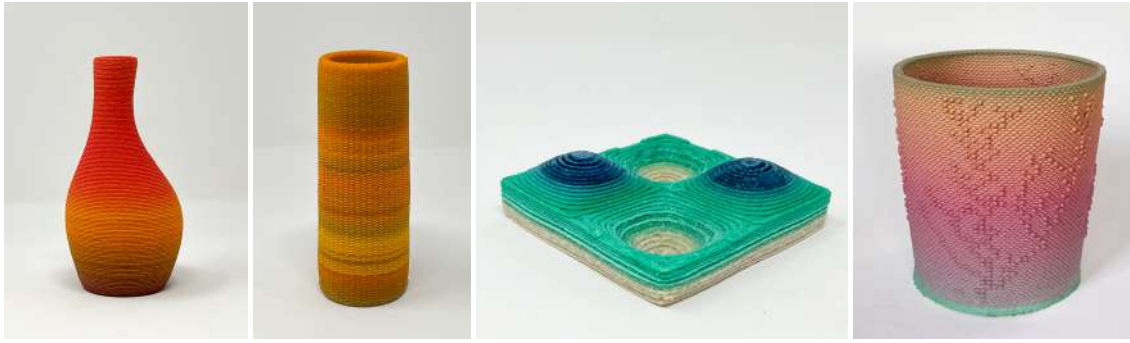


Figure 3: Colorful play-dough prints.

5 EXAMPLE ARTIFACTS

One of the most striking features of play-dough is its color. It is easy to tint play-dough an infinite variety of hues with food coloring. By adding several colors of dough to a print tube, multi-colored prints can be made. Different colors blend together as they are printed, creating an "ombre" effect. Figure 2 shows an example workflow. Color transitions are designed and visualized in software which also generates a print tube visualization based on the design. When the tube is loaded according to the guidelines, the print reflects the designed colors. Depending on the material being printed, a user may need to adjust a color diffusion parameter in the software, which defines the speed of color transitions, for accurate printing.

In addition to using software to design color effects, it is possible to work more improvisationally. A tube can be loaded with a stack of different colors and the user can wait until printing to discover the final result. We think of this technique as similar to employing a variegated yarn in knitting or weaving. An overall color pallet is chosen, but the design isn't fully known until the artifact is constructed. Figure 3 shows several colorful example prints.

We believe that techniques similar to ones investigated by other researchers for FFF machines [5][7] could be used to create higher

resolution colored prints in play-dough. The material readily absorbs dye, which could be applied when material is closer to extrusion—at or near the nozzle exit. We hope to experiment with these possibilities in future work.

6 CONCLUSION

Beyond these specific examples, we believe that, because it is so easy to make and dye your own play-dough, it is possible to create prints with a range of interesting material properties. For instance, infra-red or UV dyes could be used to embed hidden codes or tags in objects. Thermochromic inks and dyes could yield artifacts with dynamic and programmable color patterns. Similarly, conductive materials like carbon fiber or metal powders could be blended into doughs and used to make multi-functional prints. In general, this class of materials might be able to serve as an accessible sandbox for a range of exciting material experiments.

Finally, and perhaps most compellingly, we believe that the ability to employ play-dough in low-cost clay printers can open up new contexts in which to explore digital fabrication. In particular, play-dough could make the exploration of 3D printing more feasible in a range of educational contexts. Play-dough is a familiar, fun, safe, and accessible material. Moreover, simple forms can be produced very quickly. Increasing print speed significantly decreases the most important obstacle to the use of 3D printers in classrooms and other large-group educational settings. Many student projects can be printed in a relatively short time period; the bottleneck of printer access is significantly reduced.

We look forward to introducing educators, young people and DIY enthusiasts to play-dough printing and hope to see the widespread adoption of this material and others like it in the near future.

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Figure 4: Sustainability Considerations. Top: A 3D print is repaired with white glue. Bottom: Decomposing 3D printed play-dough cylinders.

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